ACM Journals

CALL FOR PAPERS

ACM Transactions on Intelligent Systems and Technology

Special Issue on LLM-Driven Agentic AI in Intelligent Systems: Theory, Technologies, and Applications

Guest Editors:

- **Xiangyu Zhao,** Associate Professor, City University of Hong Kong, Hong Kong SAR, China, xianzhao@cityu.edu.hk
- **Zhaochun Ren,** Associate Professor, Leiden University, Netherlands, z.ren@liacs.leidenuniv.nl
- **Ruocheng Guo,** Staff Research Scientist, Intuit AI Research, USA, rguo.asu@gmail.com
- **Maolin Wang,** Research Assistant Professor, City University of Hong Kong, Hong Kong SAR, China, maolin.special@gmail.com

Aims and Scope

Recent advances in Large Language Models (LLMs) have enabled a new class of agentic AI systems—systems that go beyond passive text generation and instead operate in a closed loop of perception, reasoning, and action, interacting autonomously with digital or physical environments to achieve complex goals.

While early studies have demonstrated promising capabilities of LLM-driven agents, fundamental challenges remain in their theoretical foundations, system design, reliability, controllability, and evaluation. In particular, there is a lack of principled understanding of how LLM-based agents behave as ***intelligent systems***: how they reason over long horizons, coordinate with tools and other agents, maintain internal states and memory, recover from errors, and operate robustly under real-world constraints.

This special issue aims to systematically investigate LLM-driven agentic AI from the perspective of intelligent systems research, emphasizing theory-grounded models, rigorous system design, and reproducible empirical evaluation. We particularly encourage contributions that go beyond prompt engineering or proof-of-concept demonstrations, and instead focus on principled methodologies, formal analysis, system-level insights, and measurable performance in realistic environments.

The goal of this special issue is to advance LLM-based agents from *emerging prototypes* to reliable, analyzable, and deployable intelligent systems.

Topics of Interest

We invite high-quality submissions addressing (but not limited to) the following three tightly connected aspects.

1. Theory: Foundations of Agentic Intelligence System

We welcome theoretical work that establishes formal foundations for LLM-driven agents as intelligent systems, including:

- **Formal Models of Agentic Intelligence:** Mathematical and computational models of LLM-based agent autonomy, goal formulation, decision-making under uncertainty, and agent–environment interaction.Topic Lorem ipsum dolor sit amet, consecteturs.
- **Reasoning and Planning Theory:** Theoretical analysis of reasoning paradigms (e.g., chain-of-thought, tree-of-thoughts, program synthesis) and algorithmic foundations for long-horizon planning, decomposition, and self-reflection.
- **System-Level Theories:** Formal analysis of closed-loop agent systems, including stability, convergence, error accumulation, feedback dynamics, and failure propagation.

- **Cognitive Architectures:** Theoretical frameworks integrating perception, memory, reasoning, and action, inspired by cognitive science or neuroscience.
- **Alignment, Safety, and Ethics:** Formal approaches to value alignment, safety constraints, interpretability, controllability, and ethical implications of autonomous agent behavior.

2. Technologies: Methodologies and System Design
This track focuses on principled system design and engineering methodologies for building robust and scalable agentic systems.

- **Advanced Function Calling & Tool Use:** Novel techniques and fine-tuning strategies to enhance LLMs' ability to accurately select, invoke, and chain external tools and APIs.Topic Lorem ipsum dolor sit amet, consecteturs.
- **Agent Architectures & Orchestration:** Frameworks for building robust single-agent and multi-agent systems (e.g., ReAct, AutoGen), including mechanisms for collaboration, delegation, and conflict resolution.
- **Memory & Context Management:** Technologies for efficient Retrieval-Augmented Generation (RAG), long-term episodic memory, and dynamic state management during extended interactions.
- **Evaluation & Benchmarking:** New metrics, datasets, benchmarks, simulation environments, and evaluation protocols designed specifically to assess agentic performance, including success rates in tool usage, robustness, and self-correction capabilities.

3. Applications: Domain-Specific Implementations
We invite application papers that demonstrate substantial system contributions and rigorous evaluation, rather than proof-of-concept demos.

- **Digital Intelligent Systems:** Agentic applications in software engineering and DevOps, search, recommendation and advertising, and scientific workflows and data analysis pipelines.Topic Lorem ipsum dolor sit amet, consecteturs.
- **Physical and Embodied Systems:** Agentic applications in robotics and embodied AI, smart environments and cyber-physical systems, and human–robot interaction.
- **High-Stakes Decision Systems:** Agentic applications in healthcare and clinical decision support, finance and risk analysis, and business intelligence and operations.

Important Dates

- Submissions deadline: January 31, 2027
- First-round review decisions: April 30, 2027
- Deadline for revision submissions: October 31, 2027
- Notification of final decisions: December 31, 2027
- Tentative publication: January 2028

Submission Information
Submissions must be prepared in accordance with the TIST submission guidelines (https://dl.acm.org/journal/tist/authorguidelines) and submitted via Manuscript Central (https://mc.manuscriptcentral.com/tist). All papers must be written in English and present original research that has neither been published previously nor is currently under review by any other journals or conferences. An extended version of a published conference paper may also be submitted to TIST for consideration, provided that the conference paper has not already appeared in a journal and that the submission includes significant new material, with at least 25% additional content according to ACM Policy. The review process will be conducted on a rolling basis.

For any questions and further information about this issue, please contact Maolin Wang (maolin.special@gmail.com).